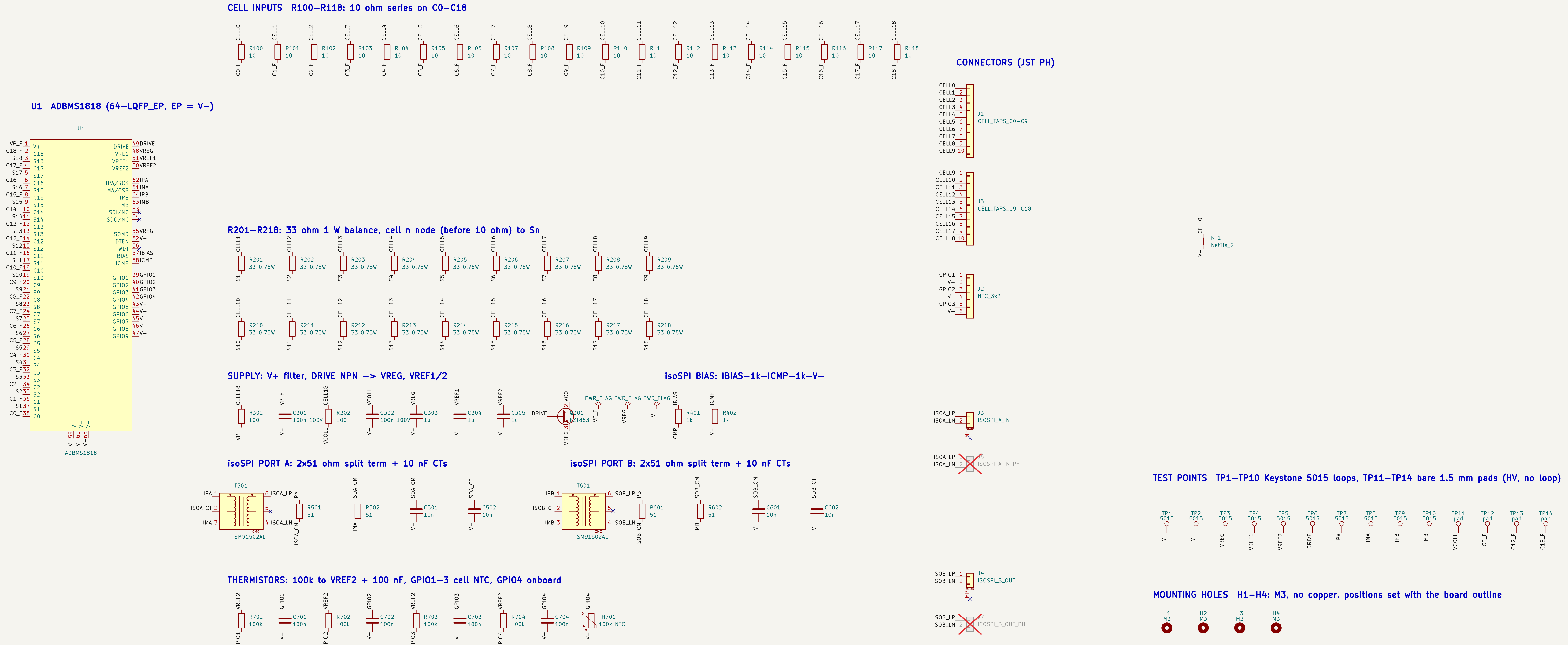
Wired with net labels: labels with the same name are one net. 18S Lean: no fuses, clamps or C−pin caps; generated from slave−18s by make_lean.py, do not edit by hand. C0 sense (CELL0, J1.1) and V− meet only at net tie NT1, which replaces F120.
ADBMS1818 Figure 112, isoSPI filtering per Fig. 104(B). See ../docs/slave−18s−design−basis.md.



CELL INPUTS R100–R118: 10 ohm series on C0–C18

U1 ADBMS1818 (64−LQFP_EP, EP = V−)

CONNECTORS (JST PH)

R201–R218: 33 ohm 1 W balance, cell n node (before 10 ohm) to Sn

SUPPLY: V+ filter, DRIVE NPN −> VREG, VREF1/2

isoSPI BIAS: IBIAS−1k−ICMP−1k−V−

isoSPI PORT A: 2x51 ohm split term + 10 nF CTs

isoSPI PORT B: 2x51 ohm split term + 10 nF CTs

THERMISTORS: 100k to VREF2 + 100 nF, GPIO1–3 cell NTC, GPIO4 onboard

TEST POINTS TP1–TP10 Keystone 5015 loops, TP11–TP14 bare 1.5 mm pads (HV, no loop)

MOUNTING HOLES H1–H4: M3, no copper, positions set with the board outline

R100 10 CELL0 C0_F
R101 10 CELL1 C1_F
R102 10 CELL2 C2_F
R103 10 CELL3 C3_F
R104 10 CELL4 C4_F
R105 10 CELL5 C5_F
R106 10 CELL6 C6_F
R107 10 CELL7 C7_F
R108 10 CELL8 C8_F
R109 10 CELL9 C9_F
R110 10 CELL10 C10_F
R111 10 CELL11 C11_F
R112 10 CELL12 C12_F
R113 10 CELL13 C13_F
R114 10 CELL14 C14_F
R115 10 CELL15 C15_F
R116 10 CELL16 C16_F
R117 10 CELL17 C17_F
R118 10 CELL18 C18_F
U1
VP_F 1 V+
C18_F 2 C18
S18 3 S18
C17_F 4 C17
S17 5 S17
C16_F 6 C16
S16 7 S16
C15_F 8 C15
S15 9 S15
C14_F 10 C14
S14 11 S14
C13_F 12 C13
S13 13 S13
C12_F 14 C12
S12 15 S12
C11_F 16 C11
S11 17 S11
C10_F 18 C10
S10 19 S10
C9_F 20 C9
S9 21 S9
C8_F 22 C8
S8 23 S8
C7_F 24 C7
S7 25 S7
C6_F 26 C6
S6 27 S6
C5_F 28 C5
S5 29 S5
C4_F 30 C4
S4 31 S4
C3_F 32 C3
S3 33 S3
C2_F 34 C2
S2 35 S2
C1_F 36 C1
S1 37 S1
C0_F 38 C0
DRIVE 49 DRIVE
VREG 48 VREG
VREF1 51 VREF1
VREF2 50 VREF2
IPA/SCK 62 IPA
IMA/CSB 61 IMA
IPB 64 IPB
IMB 63 IMB
SDI/NC 53
SDO/NC 54
ISOMD 55 VREG
DTEN 52 V−
WDT 56
IBIAS 57 IBIAS
ICMP 58 ICMP
GPIO1 39 GPIO1
GPIO2 40 GPIO2
GPIO3 41 GPIO3
GPIO4 42 GPIO4
GPIO5 43 V−
GPIO6 44 V−
GPIO7 45 V−
GPIO8 46 V−
GPIO9 47 V−
V− 59 V− 60 V− 65 V−
ADBMS1818
R201 33 0.75W CELL1 S1
R202 33 0.75W CELL2 S2
R203 33 0.75W CELL3 S3
R204 33 0.75W CELL4 S4
R205 33 0.75W CELL5 S5
R206 33 0.75W CELL6 S6
R207 33 0.75W CELL7 S7
R208 33 0.75W CELL8 S8
R209 33 0.75W CELL9 S9
R210 33 0.75W CELL10 S10
R211 33 0.75W CELL11 S11
R212 33 0.75W CELL12 S12
R213 33 0.75W CELL13 S13
R214 33 0.75W CELL14 S14
R215 33 0.75W CELL15 S15
R216 33 0.75W CELL16 S16
R217 33 0.75W CELL17 S17
R218 33 0.75W CELL18 S18
J1 CELL_TAPS_C0−C9
CELL0 1
CELL1 2
CELL2 3
CELL3 4
CELL4 5
CELL5 6
CELL6 7
CELL7 8
CELL8 9
CELL9 10
J5 CELL_TAPS_C9−C18
CELL9 1
CELL10 2
CELL11 3
CELL12 4
CELL13 5
CELL14 6
CELL15 7
CELL16 8
CELL17 9
CELL18 10
J2 NTC_3x2
GPIO1 1
V− 2
GPIO2 3
V− 4
GPIO3 5
V− 6
NT1 NetTie_2 CELL0 V−
R301 100 CELL18 VP_F
C301 100n 100V VP_F V−
R302 100 CELL18 VCOLL
C302 100n 100V VCOLL V−
C303 1u VREG V−
C304 1u VREF1 V−
C305 1u VREF2 V−
Q301 FZT853 DRIVE 1 VCOLL 2 VREG 3
PWR_FLAG VP_F
PWR_FLAG VREG
PWR_FLAG V−
R401 1k IBIAS ICMP
R402 1k ICMP V−
J3 ISOSPI_A_IN ISOA_LP 1 ISOA_LN 2 MP
J6 ISOSPI_A_IN_PH ISOA_LP 1 ISOA_LN 2
T501 SM91502AL CMC
IPA 1
ISOA_CT 2
IMA 3
6 ISOA_LP
5
4 ISOA_LN
R501 51 IPA ISOA_CM
R502 51 ISOA_CM IMA
C501 10n ISOA_CM V−
C502 10n ISOA_CT V−
T601 SM91502AL CMC
IPB 1
ISOB_CT 2
IMB 3
6 ISOB_LP
5
4 ISOB_LN
R601 51 IPB ISOB_CM
R602 51 ISOB_CM IMB
C601 10n ISOB_CM V−
C602 10n ISOB_CT V−
J4 ISOSPI_B_OUT ISOB_LP 1 ISOB_LN 2 MP
J7 ISOSPI_B_OUT_PH ISOB_LP 1 ISOB_LN 2
R701 100k VREF2 GPIO1
C701 100n GPIO1 V−
R702 100k VREF2 GPIO2
C702 100n GPIO2 V−
R703 100k VREF2 GPIO3
C703 100n GPIO3 V−
R704 100k VREF2 GPIO4
C704 100n GPIO4 V−
TH701 100k NTC GPIO4 V−
TP1 5015 V−
TP2 5015 V−
TP3 5015 VREG
TP4 5015 VREF1
TP5 5015 VREF2
TP6 5015 DRIVE
TP7 5015 IPA
TP8 5015 IMA
TP9 5015 IPB
TP10 5015 IMB
TP11 pad VCOLL
TP12 pad C6_F
TP13 pad C12_F
TP14 pad C18_F
H1 M3
H2 M3
H3 M3
H4 M3

Not for manufacture
Baseline: ADBMS1818 datasheet Figure 112 typical application

Sheet: /
File: adbms1818−slave−18s−lean.kicad_sch

Title: ADBMS1818 18S slave module − Lean (no hot−plug protection)

Size: User
Date: 2026−11−06
Rev: 0−draft
KiCad E.D.A. 10.0.6
Id: 1/1